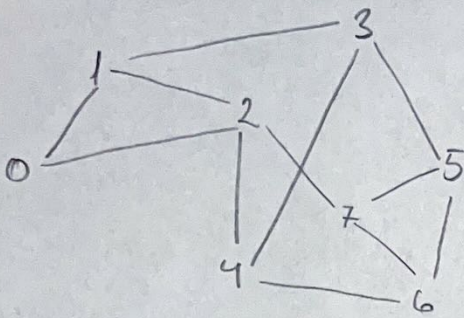


Practical work no 4

6. Given an undirected connected graph, constructs a minimal spanning tree using Prim's algorithm.



graph.txt

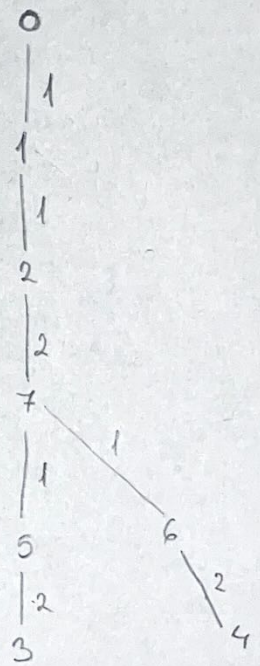
```
8 12
0 1 1
0 2 1
1 2 1
1 3 3
2 4 3
3 4 4
3 5 2
4 6 2
5 6 3
5 7 1
6 7 1
7 2 2
```

queue:

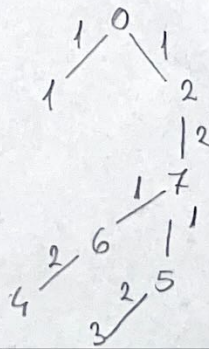
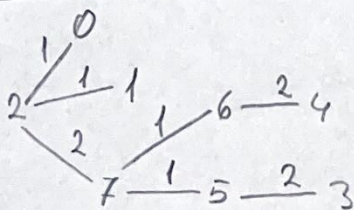
	cost	mode	
1	1	1	} initial
2	1	2	
3	3	3	} iter 1
4	3	4	
5	2	7	} iter 2
6	1	6	
7	1	5	} iter 3
8	2	3	
9	2	4	} iter 4
10	2	4	

{1,2,7,5}

initialization	selected edge	vertices	edges
iteration 1	(0,1)	{0,1}	{(0,1)}
iteration 2	(2,1)	{0,1,2}	{(0,1),(2,1)}
iteration 3	(2,7)	{0,1,2,7}	{(0,1),(2,1),(2,7)}
iteration 4	(5,7)	{0,1,2,7,5}	{(0,1),(2,1),(2,7),(5,7)}
iteration 5	(7,6)	{0,1,2,5,7,6}	{(0,1),(2,1),(2,7),(5,7),(7,6)}
iteration 6	(3,5)	{0,1,2,3,5,6,7}	{(0,1),(2,1),(2,7),(5,7),(7,6),(3,5)}
iteration 7	(4,6)	{0,1,2,3,4,5,6,7}	{(0,1),(2,1),(2,7),(5,7),(7,6),(3,5),(4,6)}



⇒ the cost is 10



other MST